



AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
International Credit Hour Programs (ICHEP)

ECE411
IC Technology

Major Task Report

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1 6T SRAM

Part A in this project focuses on the design and verification of a circuit layout extracted from a symbolic layout, which can be shown in figure 1.

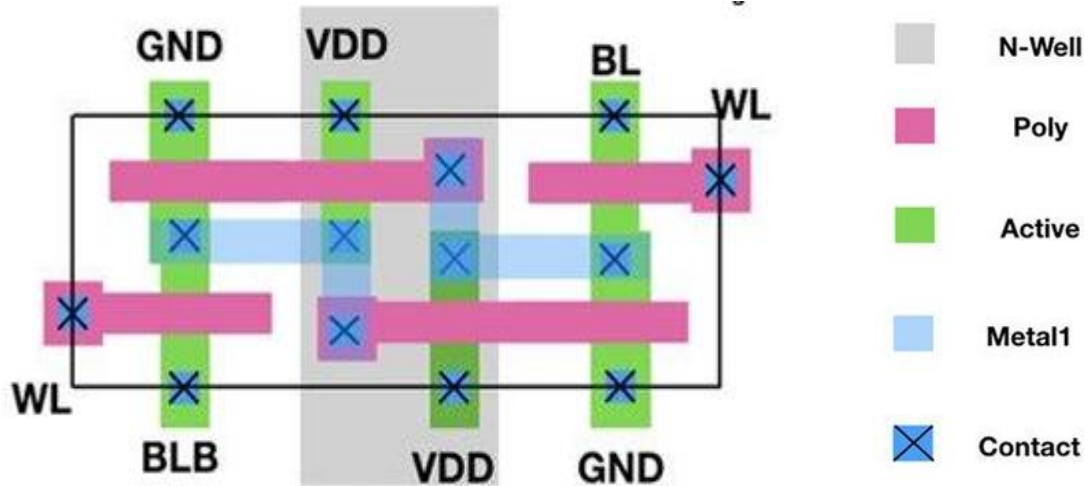


Figure 1 shows the symbolic layout which will be the main circuit.

For this project, the TSMC 65nm CMOS technology node is utilized. At this scale, parasitic effects and layout dependent phenomena become more pronounced, emphasizing the necessity for precise layout practices and post layout verification. The process rules for the 65nm node are stricter, making it crucial to ensure compliance with the Design Rule Check (DRC). To verify the layout, Calibre is used. It ensures that the layout adheres to the required physical design rules and checks for functional equivalence between the layout and schematic through two critical processes: Design Rule Checking (DRC) and Layout Versus Schematic (LVS) verification. These checks are essential for confirming that the layout is both manufacturable and functionally correct.

1.1 The 6T SRAM Cell Architecture

At its core, the memory cell utilizes six transistors to store and access a single bit of data. The storage mechanism consists of two cross-coupled CMOS inverters, creating a bistable latch that holds its state as long as power is applied. Access to the internal storage nodes (Q and QB) is controlled by two NMOS pass transistors, which are driven by the Wordline (WL) and interface with the complementary Bitlines (BL and BLB). The operational stability of this architecture relies heavily on the precise relative sizing of the pull-up, pull-down, and access transistors.

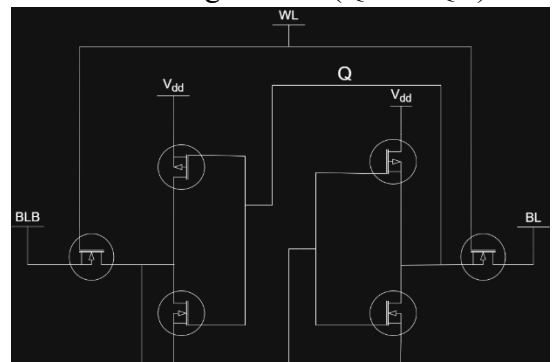


Figure 2 Conceptual schematic of the 6T SRAM cell

1.1.1 Read Operation & CR Constraint

A read operation begins by precharging both complementary bitlines (BL and BLB) to VDD. Once the precharge phase concludes, the wordline (WL) is asserted, activating the access transistors and connecting the internal storage nodes (Q and QB) to the bitlines. The bitline connected to the internal node storing a logic '0' will begin to discharge to ground through its corresponding access transistor and pull-down NMOS. During this discharge, a temporary voltage bump occurs at the '0' node due to the voltage divider formed between the access and pull-down transistors.

To ensure read stability and prevent a destructive read (Read Upset), this voltage bump must remain safely below the switching threshold of the opposite inverter. This constraint is governed by the Cell Ratio (CR), defined as the ratio of the width/length sizing between the pull-down NMOS and the access NMOS. A strictly maintained CR guarantees that the pull-down transistor is strong enough to keep the internal node grounded during a read access.

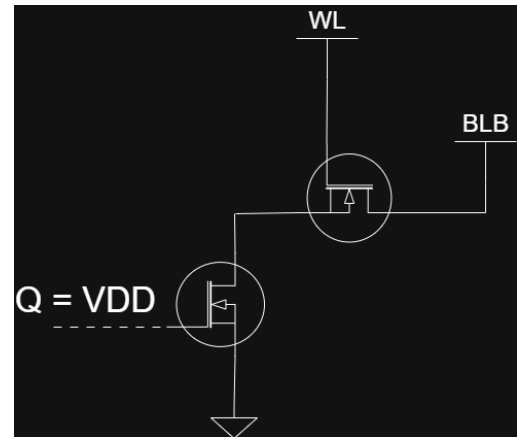


Figure 3 Equivalent circuit during a read operation, illustrating the voltage divider formed by the access and pull-down transistors

1.1.2 The Write Operation & PR Constraint

A write operation requires forcing the internal storage nodes to a new state. The bitlines are driven to complementary logic levels (e.g., BL = VDD and BLB = 0V for a Write ('0')) by the write drivers. Upon asserting the wordline, the access transistors open, causing a physical contention between the access transistor driving the '0' and the internal pull-up PMOS currently holding the '1' state.

For a successful write, the access transistor must overpower the pull-up PMOS to drag the '1' node's voltage low enough to flip the cross-coupled inverters. This writeability is governed by the **Pull-Up Ratio (PR)**, defined as the ratio of the width/length sizing between the pull-up PMOS and the access NMOS. The PR must be kept sufficiently low to ensure the access transistor can successfully discharge the node and force the state change.

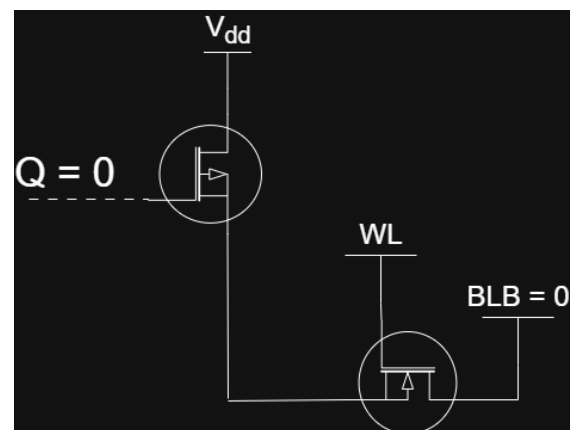


Figure 4 Equivalent circuit during a write operation, highlighting the contention between the pull-up PMOS and the access transistor

1.2 Schematic Drawing and Sizing

With the theoretical constraints established, the 6T SRAM cell was implemented at the transistor level using Cadence Virtuoso. To minimize the overall area footprint of the memory array, the minimum channel length allowed by the TSMC 65nm design kit ($L = 60\text{nm}$) was utilized for all six transistors. The widths of the transistors were then systematically derived to satisfy the Cell Ratio (CR) and Pull-Up Ratio (PR) constraints.

1.2.1 Transistor Sizing and Ratio Calculations

The sizing strategy began by assigning the access transistors to a specific width. The **access transistors** were sized with a width of **300nm**. Based on this baseline, the pull-down and pull-up transistors were sized as follows:

- **Read Stability Sizing (CR):**

To satisfy the read constraint, a Cell Ratio of 1.2 was selected. This sets the width of **the pull-down NMOS to 360nm**. At this ratio, during a read operation, the voltage Rise (the temporary bump at the grounded storage node) is limited to approximately 0.4V. Because the switching threshold of the opposing internal inverter sits roughly at half of the supply voltage $\frac{V_{DD}}{2} = 0.6$, this 0.4V bump provides a safe operational margin of 0.2V. This margin guarantees that the cell avoids a destructive read upset.

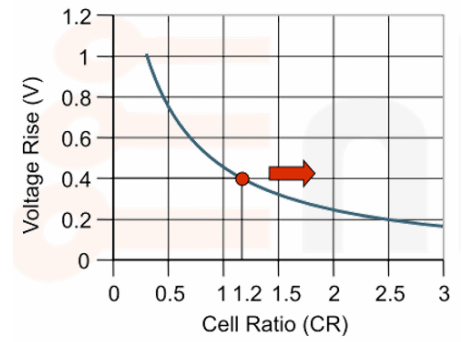


Figure 5 Voltage rise versus Cell Ratio (CR)

- **Write Stability Sizing (PR):**

To satisfy the write constraint, a Pull-Up Ratio of 0.8 was selected. This sets the width of the pull-up PMOS to 240nm. By keeping the PR strictly below 1.0, the access transistor remains sufficiently stronger than the pull-up PMOS, ensuring it can successfully overpower the PMOS to discharge a logic '1' node and successfully flip the cell's state during a write operation.

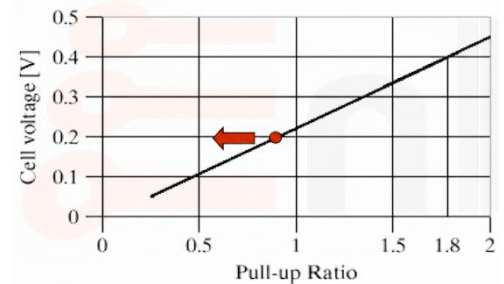


Figure 6 cell voltage versus Pull-Up Ratio (PR)

1.2.2 Cadence Virtuoso Schematic

With the dimensions calculated, the cell was fully routed in the Cadence schematic editor as in the following figure.

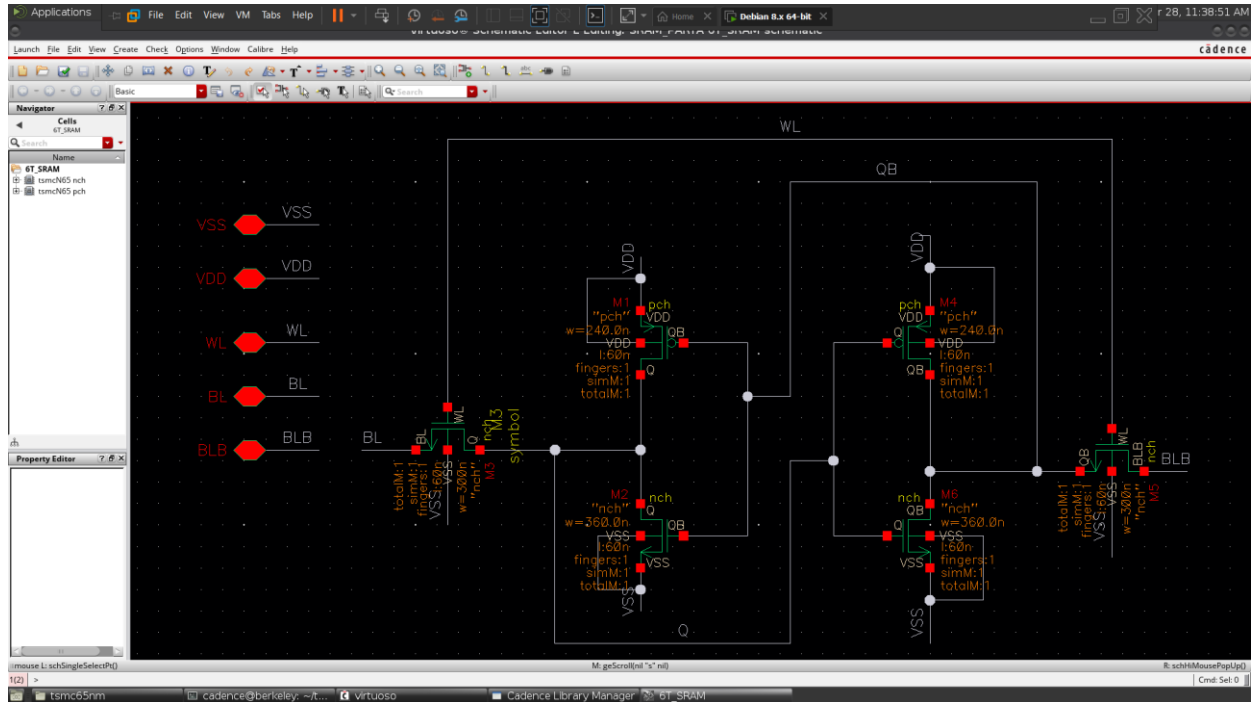


Figure 7 Complete 6T SRAM cell schematic implemented in Cadence Virtuoso

1.3 Schematic Verification

To thoroughly verify the mathematically sized SRAM cell's functionality, a transient testbench was designed in Cadence ADE. The simulation forces the cell through a continuous **Read** → **Hold** → **Write** sequence to prove stability across all operating modes.

1.3.1 Testbench Architecture

The testbench relies on two critical peripheral circuits to stimulate the SRAM cell, mainly they control the **BL/BLB**.

1. **The Write Drivers:** Two NMOS pass-transistors are controlled by a **WENABLE** signal. These drive the differential input data (Data and it's inverse) onto the bitlines during a write operation only.

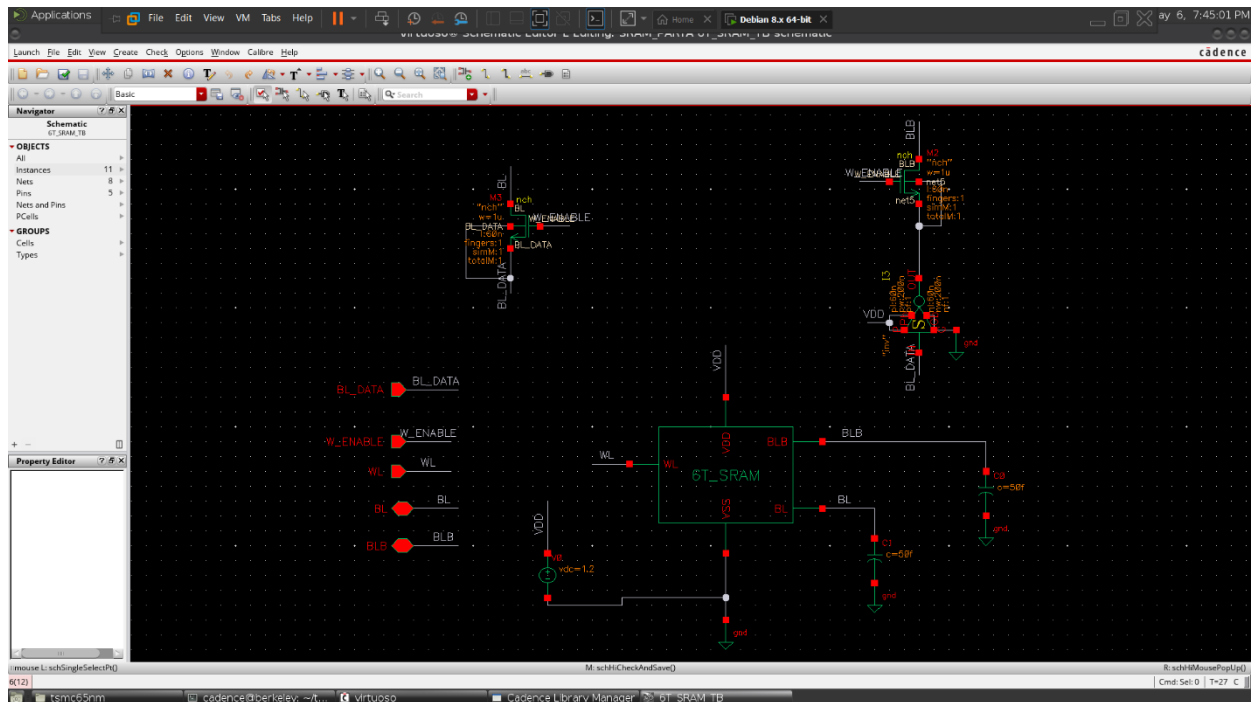


Figure 8 Schematic of the SRAM testbench

1.3.2 Transient Simulation Results

The transient simulation successfully executed the multi-cycle sequence, confirming the mathematical sizing of the cell:

1. **Read Phase:** With the bitlines initially precharged, WL was asserted to read the logic '0' stored in the cell ($Q = 0V$, $QB = 1.2V$). As the bitline discharged to ground, a temporary read-disturb voltage bump was observed on the internal grounded node (Q). The bump peaked at approximately 220mV, which remains safely below the switching threshold of the opposing inverter. This confirms the Cell Ratio (CR) sizing successfully prevented a destructive read upset.
2. **Hold Phase:** WL was driven low, isolating the cell and allowing the internal inverters to safely hold their latched state. Simultaneously, the write drivers were enabled to pre-condition the bitlines with the upcoming data. The NMOS pass-transistors demonstrated their physical limitations here: the driver passing a '1' to BL suffered a threshold voltage drop, maxing out at ~945mV (a "weak 1"), while the driver passing a '0' to BLB pulled the line down cleanly to ~27mV (a "strong 0").
3. **Write Phase:** With the bitlines fully loaded with the new data, WL was asserted. The strong 0V on BLB successfully overpowered the internal pull-up PMOS, dragging QB low and forcing the cell to flawlessly flip to the new state ($Q = 1.2V$).



Figure 9 Transient simulation of the multi-cycle testbench.

- **Stimulus Used:**

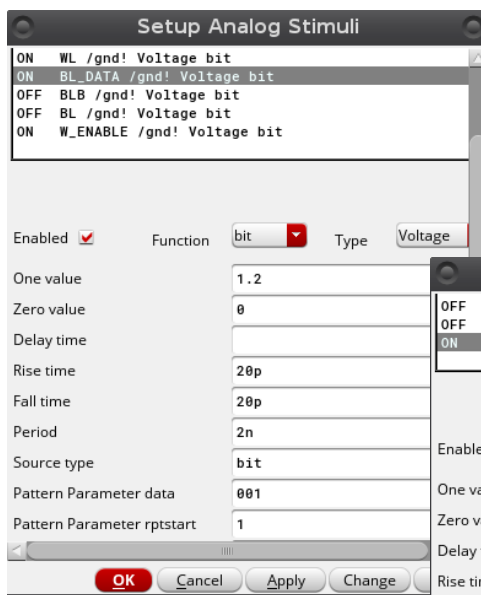


Figure 12 Setup Analog Stimuli configuration for the BL_DATA

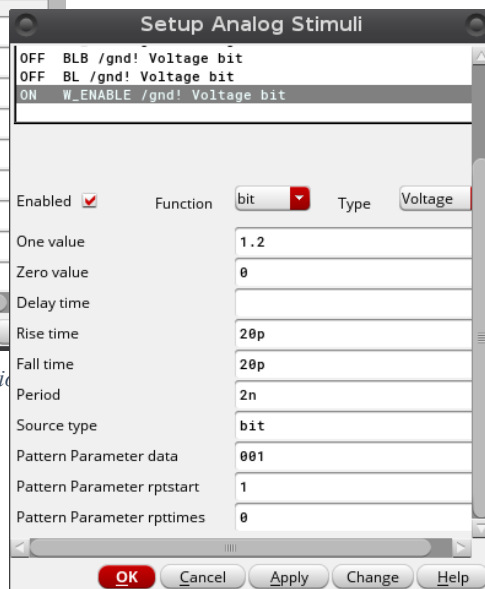


Figure 11 Setup Analog Stimuli configuration for the W_ENABLE

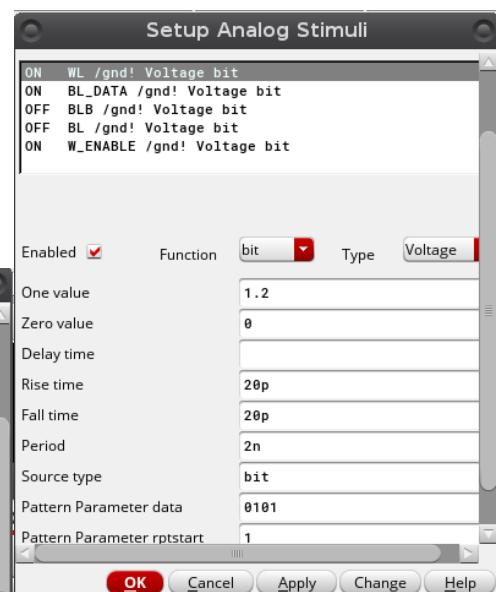


Figure 10 Setup Analog Stimuli configuration for the WL

- Initial Conditions used:

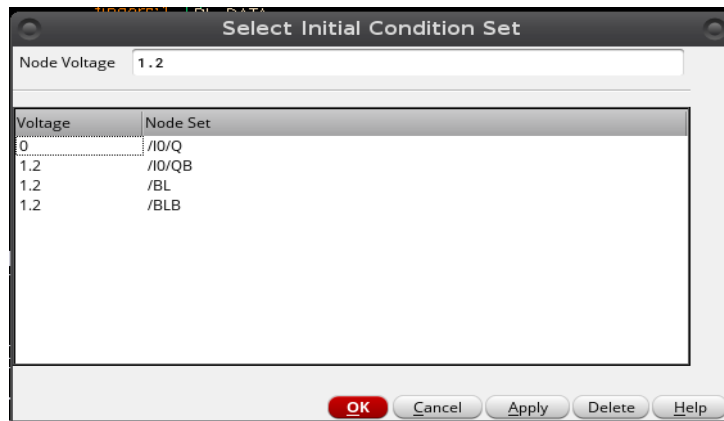


Figure 13 Initial Condition (IC) setup forcing the cell to start with a stored '0' ($Q=0V$, $QB=1.2V$) and both bitlines 1.2V.

1.4 Physical Implementation & Parasitic Extraction

With the schematic baseline verified, the 6T SRAM cell was transitioned to a physical layout using the TSMC 65nm design rules. The objective of this phase is to create an area-efficient layout, verify its physical connectivity, and extract the parasitic capacitances and resistances (RC) for post-layout analysis.

1.4.1 Layout and Physical Verification

The layout was drawn in Cadence Virtuoso. Careful attention was paid to routing the VDD and GND rails, as well as minimizing the distance between the cross-coupled inverters to reduce internal node capacitance.

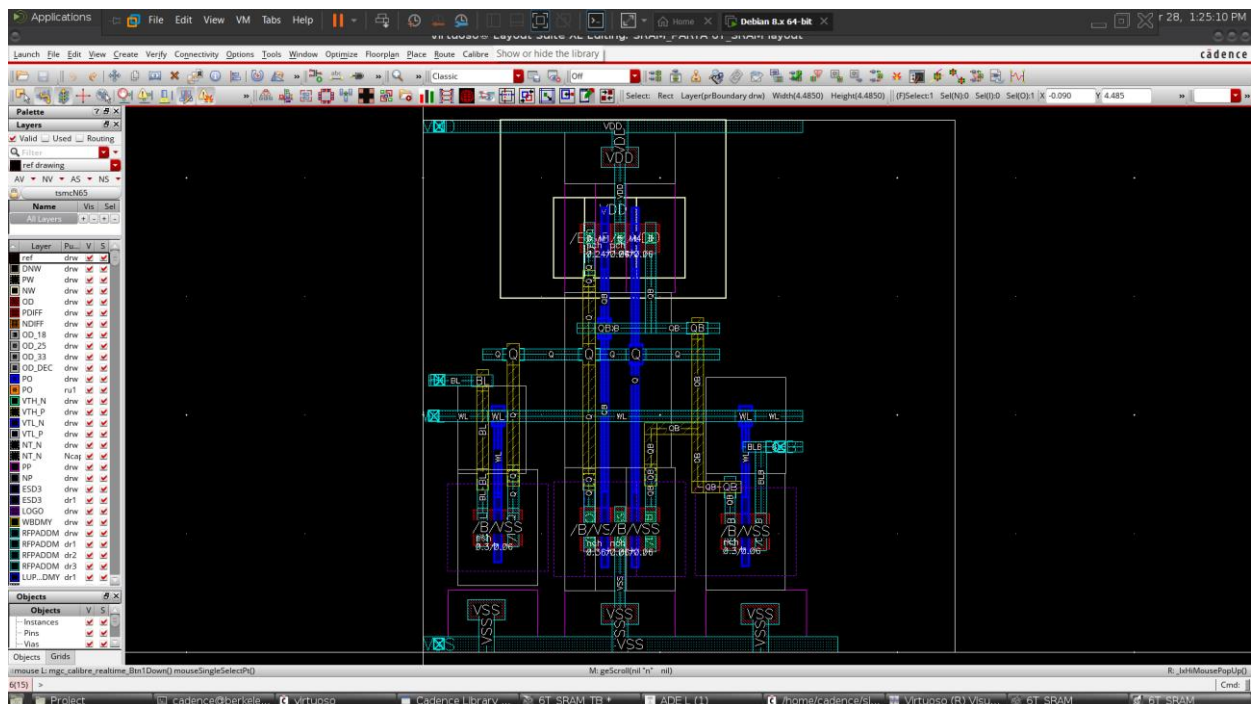


Figure 14 shows the layout implemented on cadence.

1.4.2 Design Rule Check

To ensure manufacturability and adherence to the foundry's physical constraints, a Design Rule Check (DRC) was performed using Mentor Graphics Calibre. The layout successfully passed all DRC checks with zero violations, confirming proper spacing, minimum widths, and enclosure rules were strictly followed.

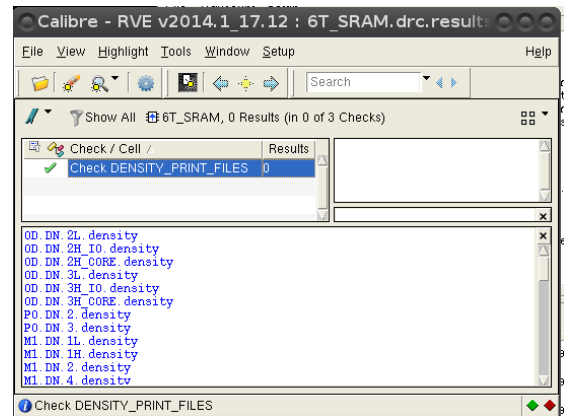


Figure 15 Calibre DRC summary report confirming zero layout violations

1.4.3 Layout Versus Schematic

Layout Versus Schematic (LVS) verification was executed to guarantee that the physical geometry perfectly matched the golden schematic netlist. The Calibre LVS report confirmed a clean match, verifying that all transistor dimensions, node connectivity, and terminal definitions were correctly translated from the schematic to the layout.

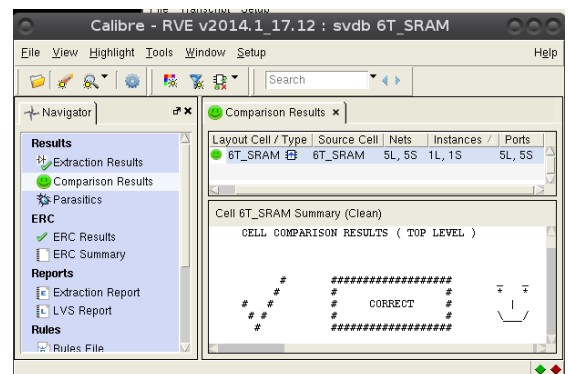


Figure 16 Calibre LVS summary report displaying the "Correct" verification

1.4.4 PEX Extraction

Calibre Parasitic Extraction (PEX) was performed to generate an extracted view containing all layout parasitics.

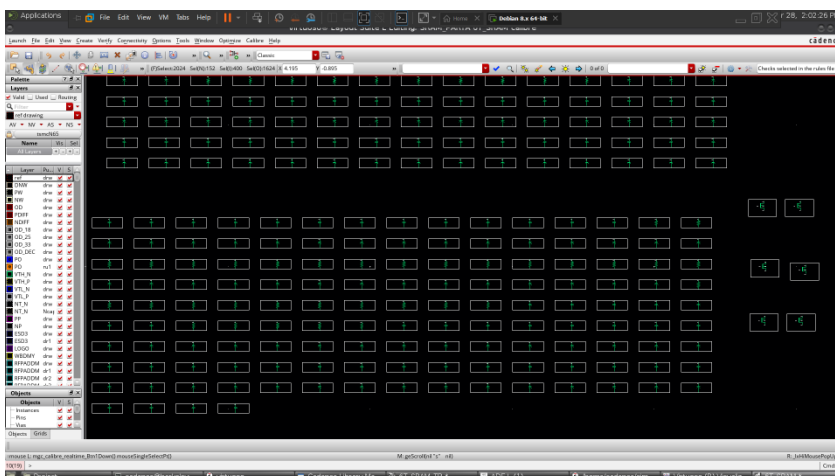


Figure 18 PEX Results

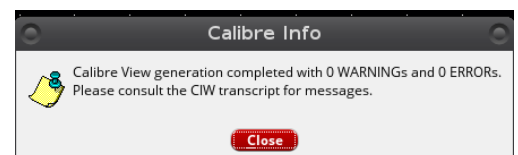


Figure 17 Successful completion of the Calibre Parasitic Extraction

1.5 Post-Layout Verification

To accurately assess the impact of the physical layout, the multi-cycle testbench was re-simulated using the Calibre PEX extracted netlist.

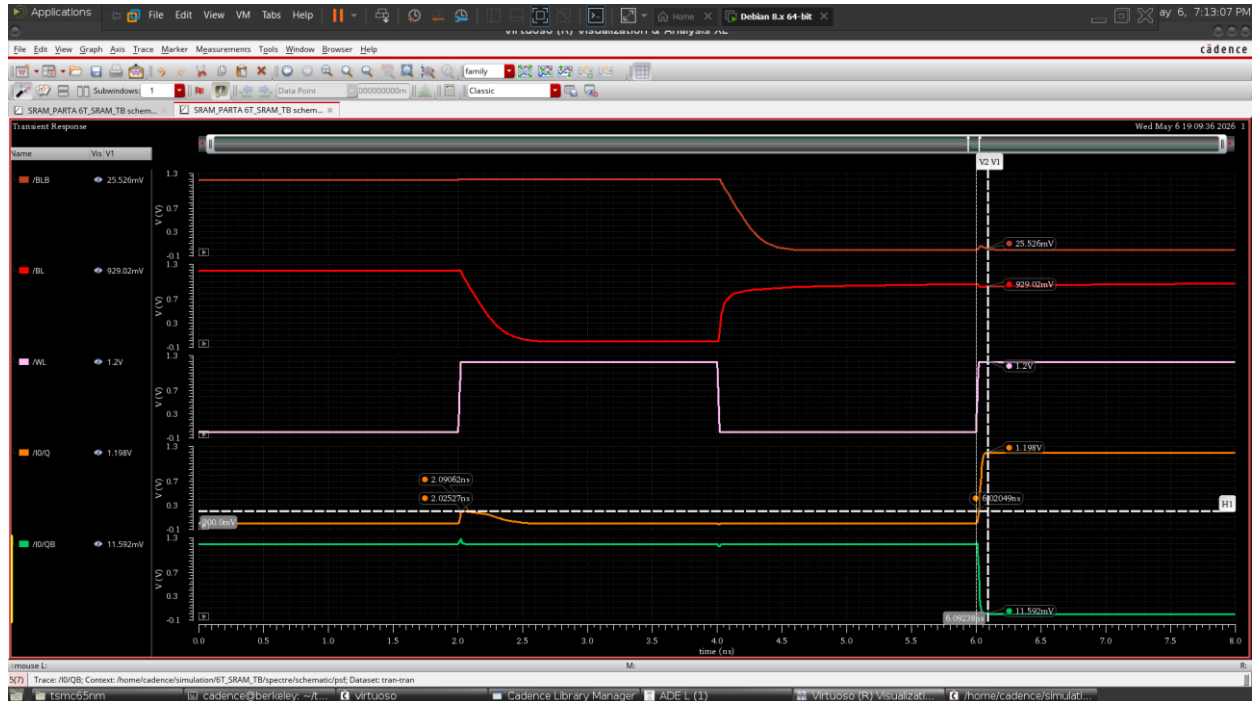


Figure 19 Output Run with PEX.

The write delay of the isolated internal storage node (Q) was measured and compared against the ideal schematic baseline:

- **Schematic Baseline Delay:** In the ideal schematic, lacking physical routing, the transition time for node Q to fall to 0V during a Write '1' operation was measured at approximately **50ps**.
- **Extracted Layout (PEX) Delay:** Following physical layout, the PEX simulation revealed that the same transition required approximately **92ps**.

2.1.1 Read Operation & Internal Node Connection

8T architecture utilizes a single-ended read port, the polarity of the read operation is entirely dependent on which internal storage node is connected to the gate of the Read Driver. The read stack essentially acts as an inverting buffer:

- **Inverting Read (Gate connected to Q):** If the cell stores a logic '1', Read Driver NMOS is turned ON. During a read operation, the RBL finds a path to ground and discharges to '0'. Conversely, storing a '0' leaves M7 OFF, keeping RBL at '1'. **In this configuration, which is utilized in this design, the RBL consistently outputs the inverted state of the stored value.**

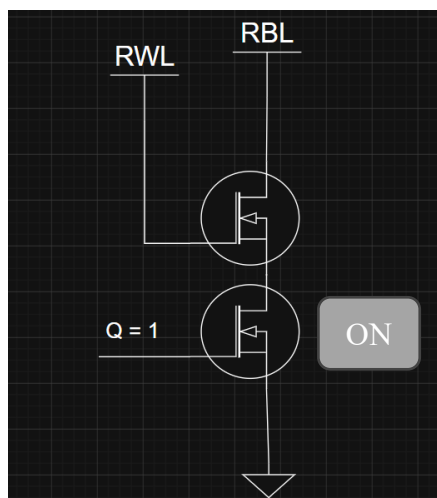


Figure 23 Shows an inverting read operation when storing a data '1'

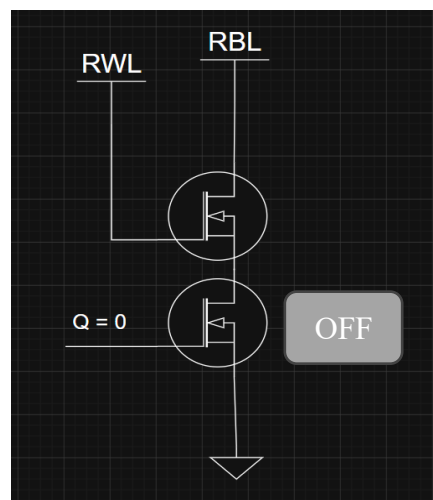


Figure 22 Shows an inverting read operation when storing a data '0'

- **Non-Inverting Read (Gate connected to BQ):** If the gate of Read Driver NMOS is instead connected to the complementary node QB, the logic is reversed. Storing a '1' (QB = 0) keeps Read Driver NMOS OFF, allowing RBL to remain at a logic '1', perfectly matching the stored data in node Q.

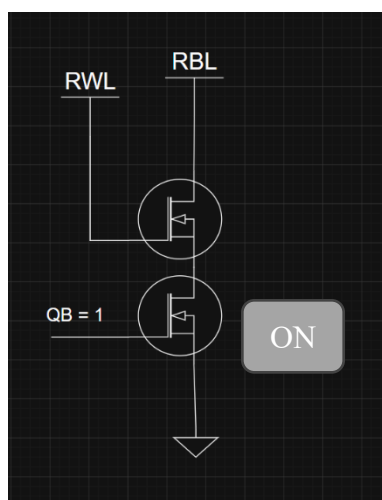


Figure 25 Shows a non-inverting read operation when storing a complementary '1'

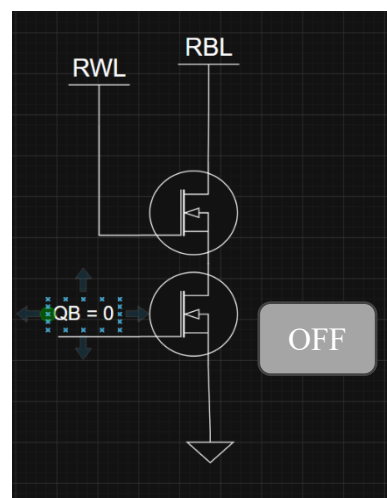


Figure 24 Shows a non-inverting read operation when storing a complementary '0'

2.2 Schematic Drawing and Sizing

The transition from the architectural concept to the transistor-level schematic was driven by the need for a strict, controlled comparison against the baseline 6T cell.

2.2.1 6T Core Reuse & Read Port Sizing

The entire core of the SRAM cell comprising the cross-coupled inverters and the write access transistors was ported directly from the 6T design without any dimension changes. Retaining these exact widths guarantees that the Writability (Pull-Up Ratio) and fundamental hold stability remain identical to the baseline.

For the newly introduced 2T read port, the sizing strategy fundamentally shifts. Because the read stack is completely isolated from the internal storage nodes, the traditional Cell Ratio (CR) constraint is obsolete; there is zero risk of a read-upset regardless of how large or strong the read transistors are.

Freed from the need to balance transistor strengths to prevent a read-upset, Read Driver NMOS and a Read Access NMOS were implemented using the absolute minimum default dimensions allowed by the technology node **W = 200nm**, **L = 60nm**.

2.2.2 Cadence Virtuoso Schematic

With the core reused and the minimum-sized read port attached, the 8T cell was fully routed in the Cadence schematic editor.

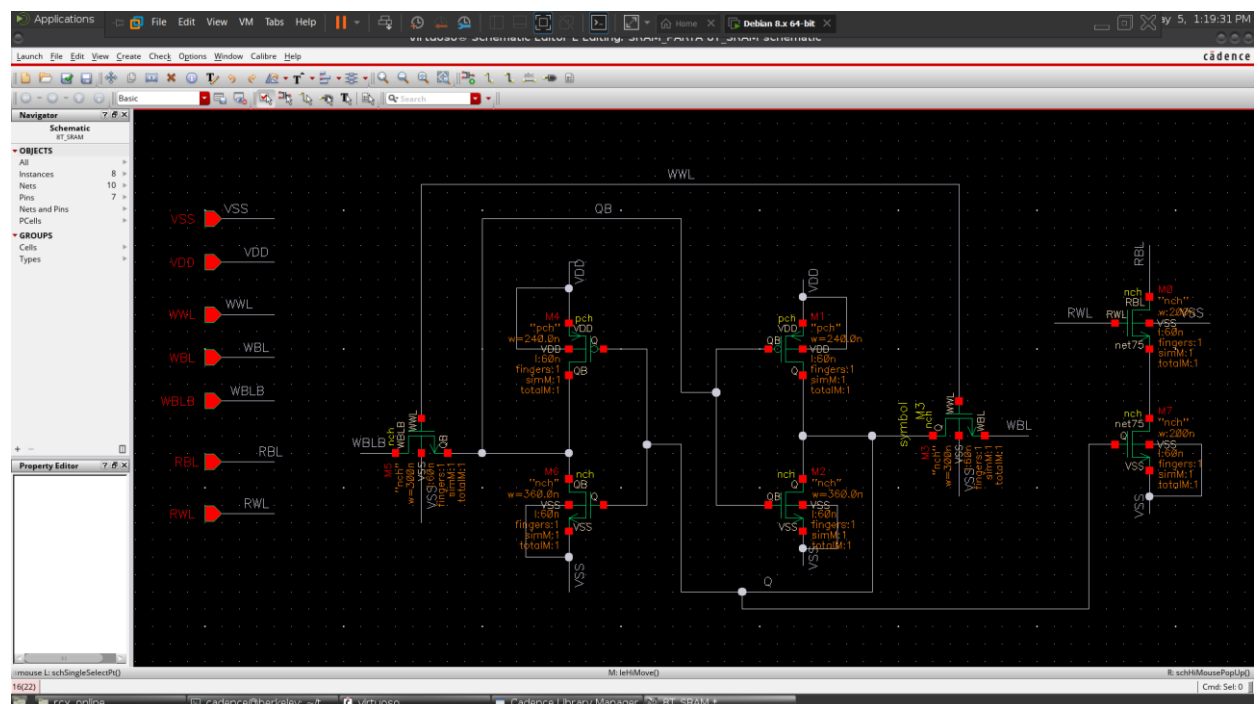


Figure 26 Complete 8T SRAM cell schematic implemented in Cadence Virtuoso

2.3 Schematic Verification

To rigorously verify the functionality and stability of the 8T SRAM cell, a multi-cycle transient testbench was created to stimulate the newly decoupled read and write ports independently. The simulation executes a continuous Write '1' $\rightarrow$ Read '1' $\rightarrow$ Write '0' sequence to prove both functionality and read-disturb immunity.

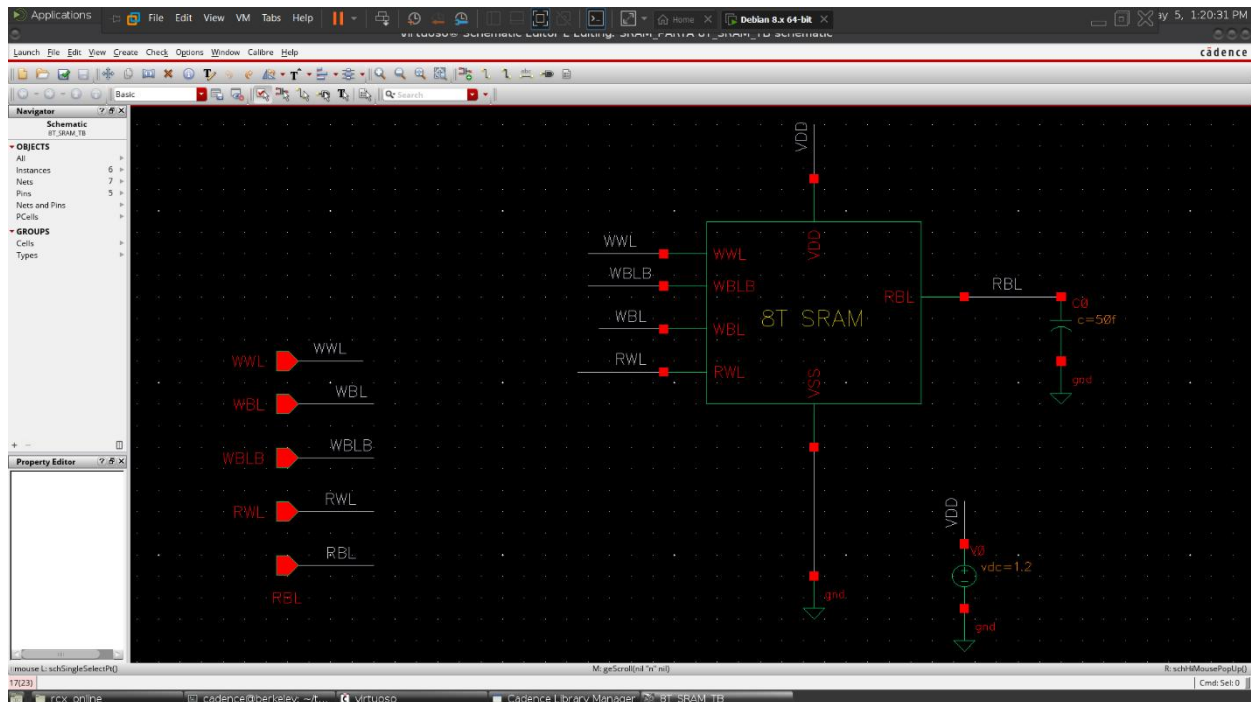


Figure 27 Schematic of the 8T SRAM testbench

2.3.1 Transient Simulation Results

The transient simulation successfully executed the multi-cycle sequence, providing definitive proof of the 8T architecture's primary advantage:

1. **Write Phase:** The write drivers successfully forced differential voltages onto the write bitlines (WBL = 1.2V, WBLB = 0V). Upon asserting WWL, the access transistors flipped the internal nodes to store logic '1' (Q = 1.2V, QB = 0V).
2. **Read Phase:** With RBL fully charged to 1.2V, RWL was asserted. Because the cell stored a logic '1' in node Q, the internal Read Driver was turned ON, successfully discharging the RBL to output a logic '0'. Crucially, during this active discharge, the internal Q node was observed to remain perfectly flat at 1.2V. Unlike the 6T baseline which suffered ~220mV read-disturb voltage bump the 8T read port's complete electrical isolation, effectively eliminating the risk of a destructive read upset.
3. **Write Phase:** WWL was asserted while the write drivers NMOSs applied inverted data (WBL = 0V, WBLB = 1.2V). the access transistors flipped the internal nodes to store logic '0' (Q = 0V, QB = 1.2V).

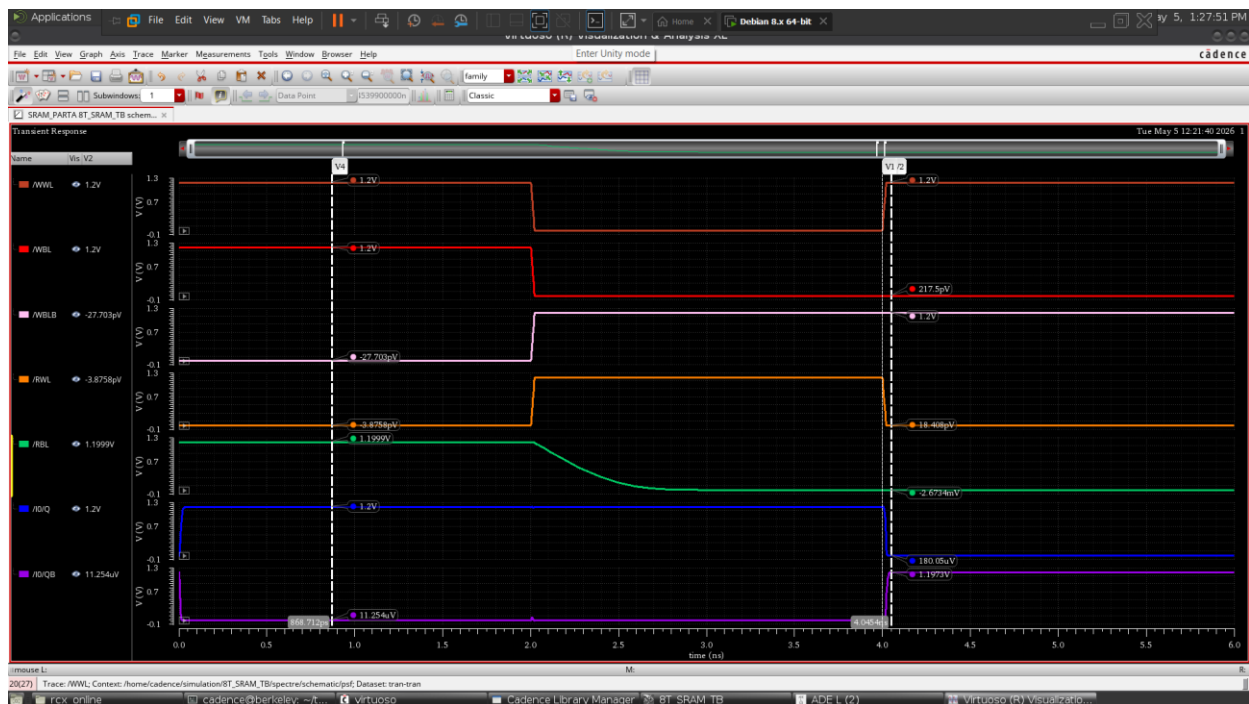


Figure 28 Transient simulation of the 8T SRAM cell showcasing the Write '1' → Read '1' → Write '0' sequence.

- Stimulus Used:

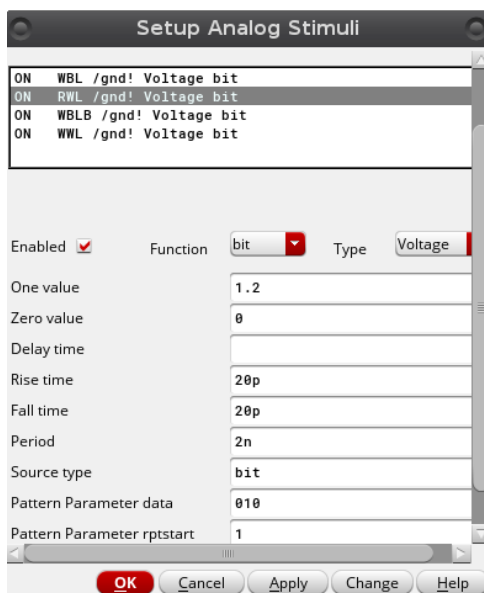


Figure 30 Setup Analog Stimuli configuration for the RWL

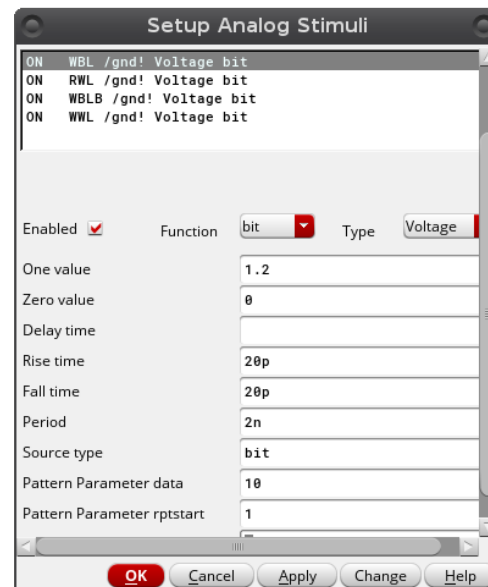


Figure 29 Setup Analog Stimuli configuration for the WBL

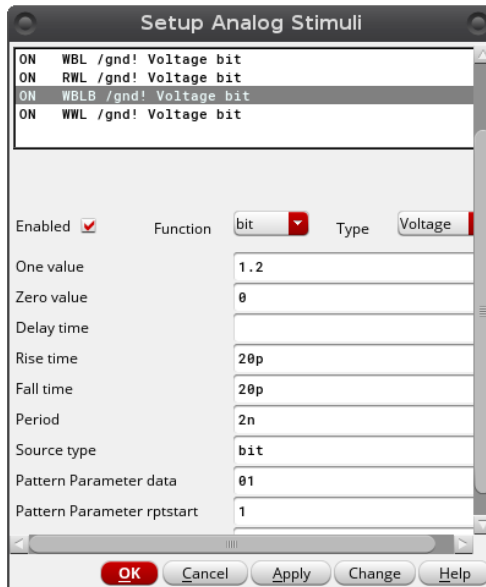


Figure 32 Setup Analog Stimuli configuration for the WBLB

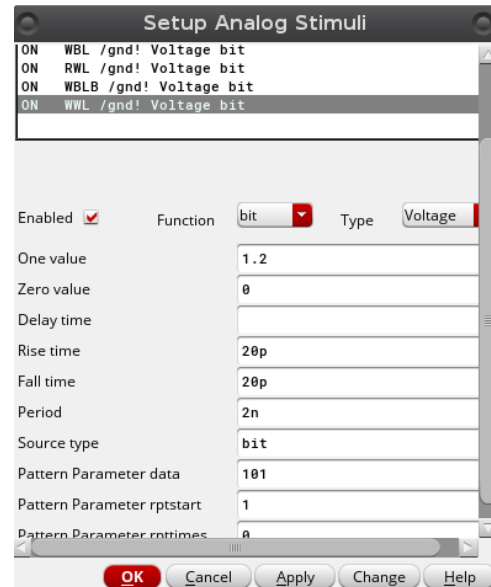


Figure 31 Setup Analog Stimuli configuration for the WWL

- Initial Conditions used:

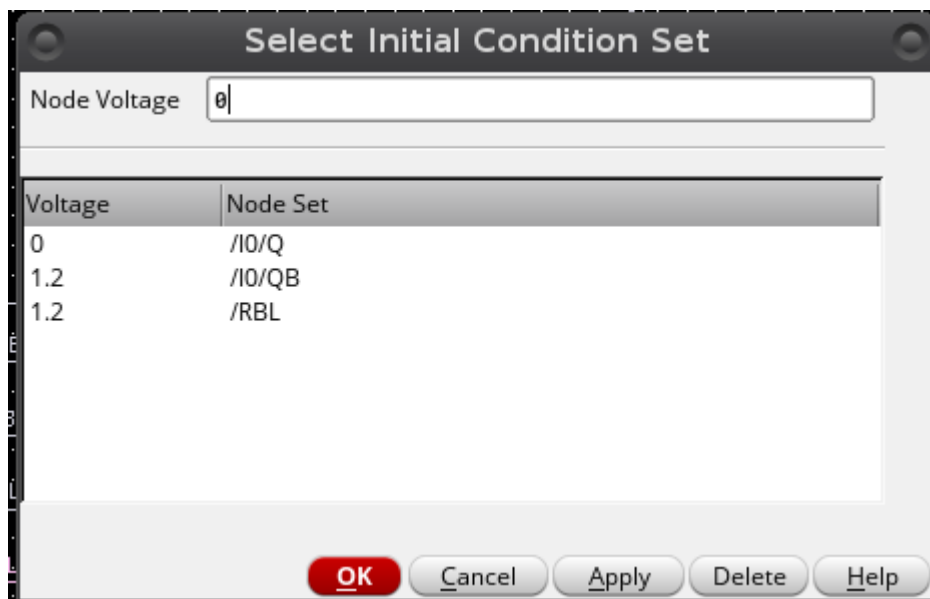


Figure 33 Initial Condition (IC) setup forcing the cell to start with a stored '0' ($Q=0V$, $QB=1.2V$) and read-bitlines $1.2V$.

2.4 Physical Implementation & Parasitic Extraction

With the schematic baseline verified, the 8T SRAM cell was transitioned to a physical layout using the TSMC 65nm design rules. The objective of this phase is to create an area-efficient layout, verify its physical connectivity, and extract the parasitic capacitances and resistances (RC) for post-layout analysis.

2.4.1 Layout and Physical Verification

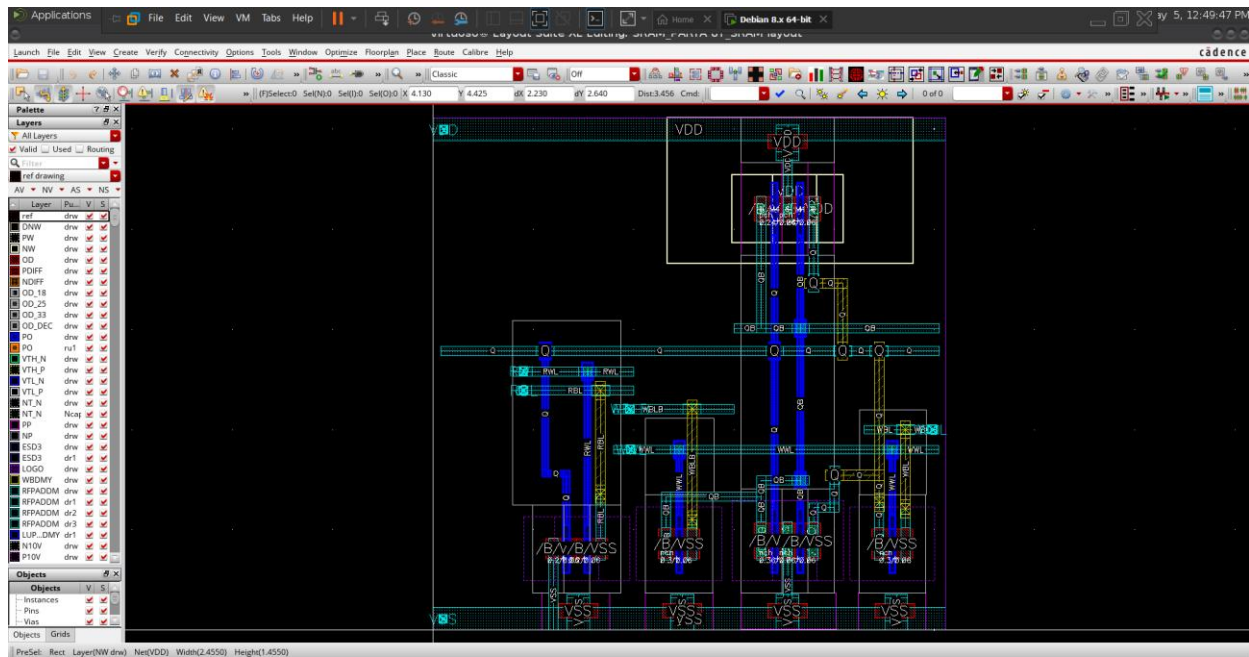


Figure 34 Physical layout of the 8T SRAM cell

2.4.2 Design Rule Check

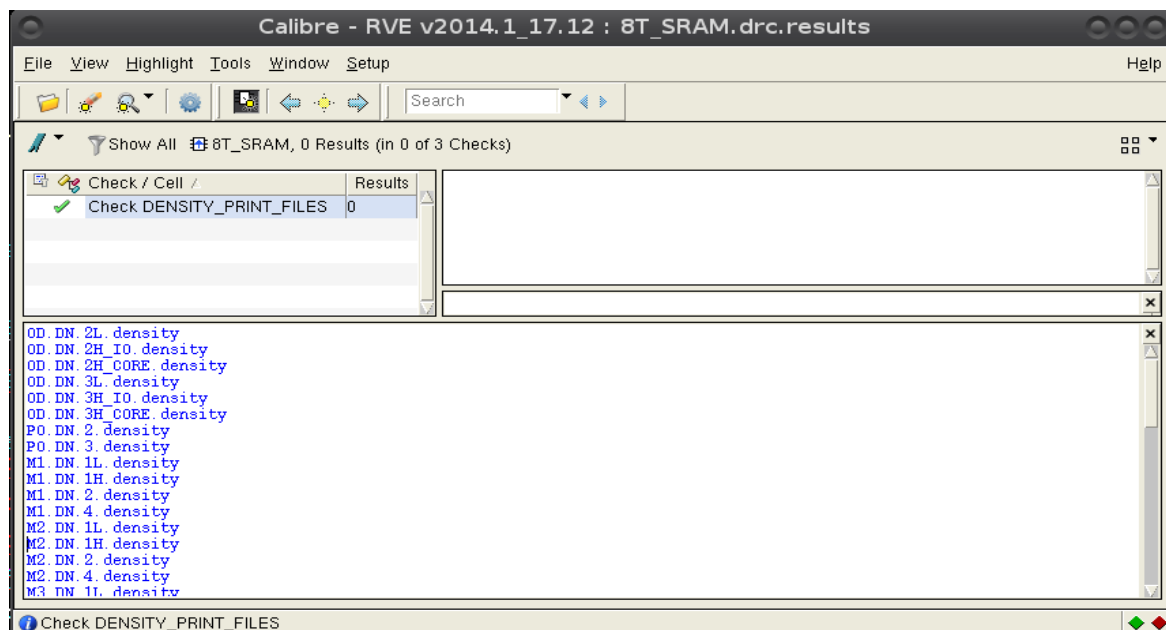


Figure 35 Calibre DRC summary report confirming zero layout violations

2.4.3 Layout Versus Schematic

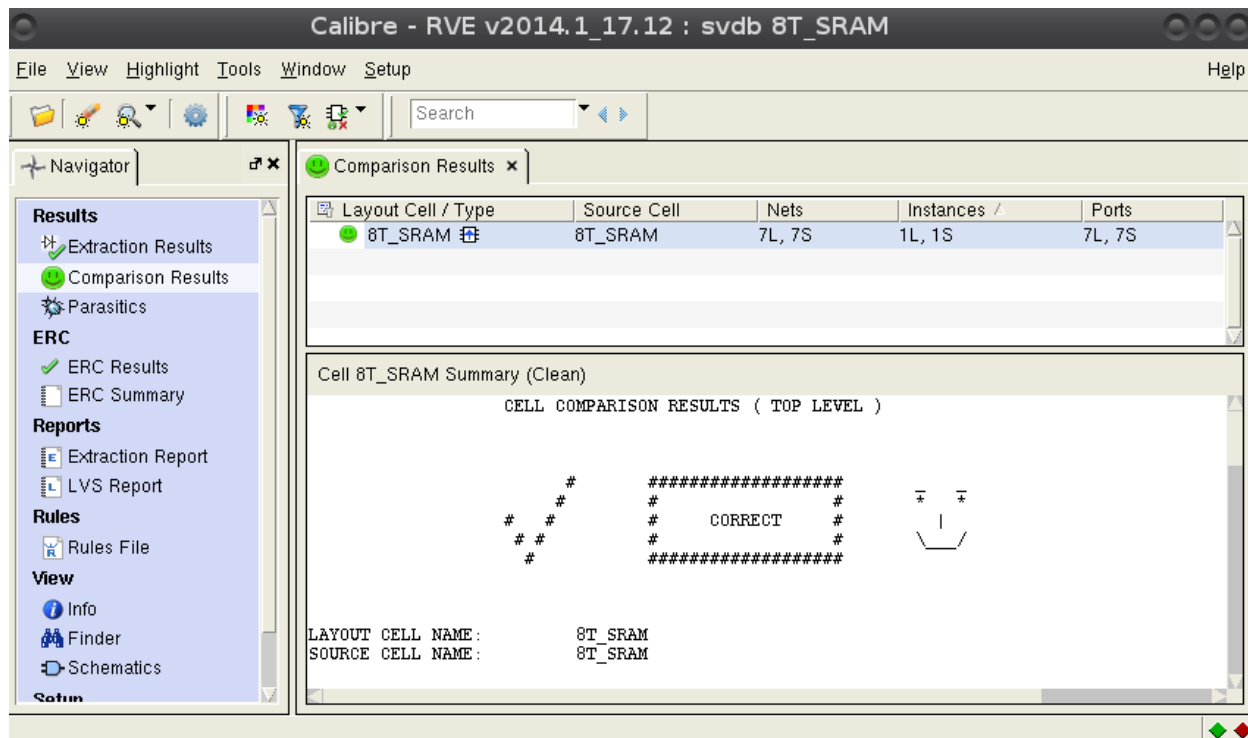


Figure 36 Calibre LVS summary report displaying the "Correct" verification

2.4.4 PEX Extraction

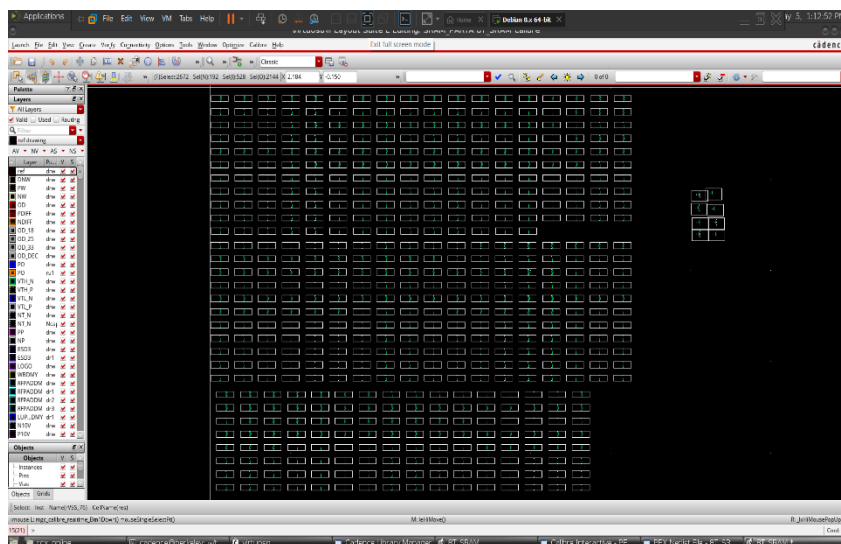


Figure 38 PEX Results

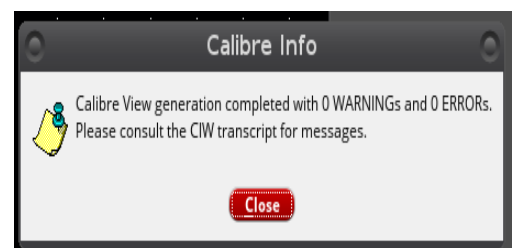


Figure 37 Successful completion of the Calibre Parasitic Extraction

2.5 Post-Layout Verification

To accurately assess the impact of the physical layout, the multi-cycle testbench was re-simulated using the Calibre PEX extracted netlist.

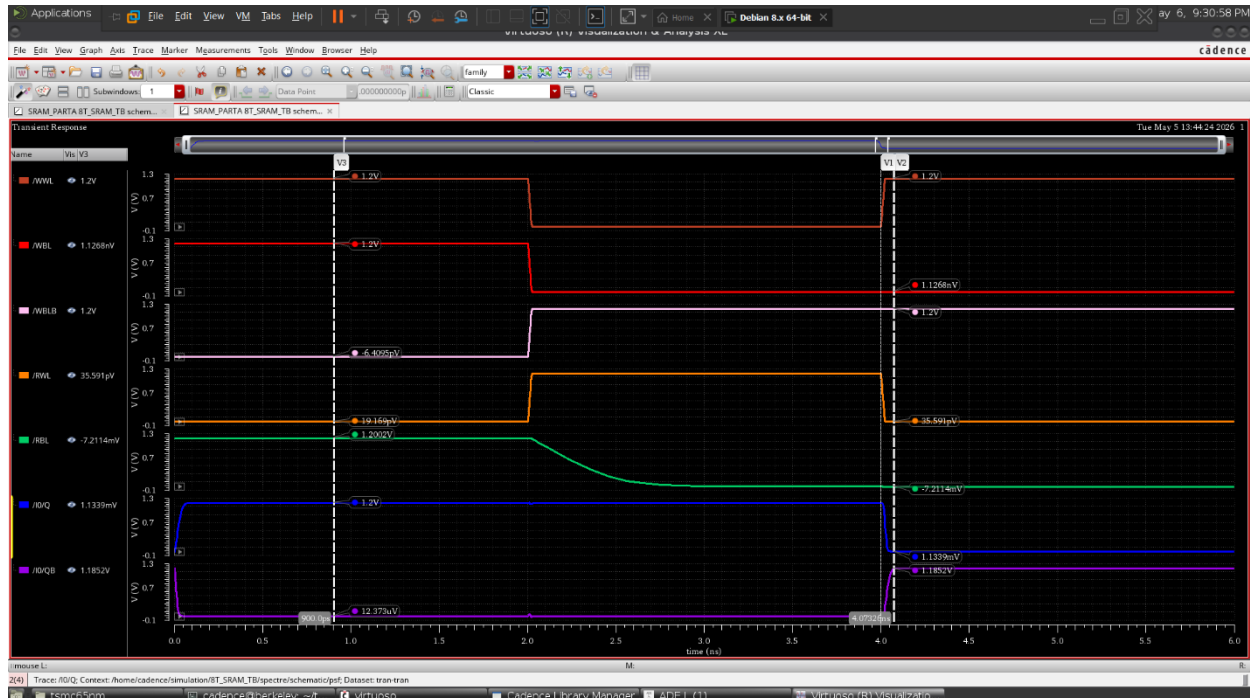


Figure 39 Output Run with PEX.

The write delay of the isolated internal storage node (Q) was measured and compared against the ideal schematic baseline:

- **Schematic Baseline Delay:** In the ideal schematic, lacking physical routing, the transition time for node Q to fall to 0V during a Write '0' operation was measured at approximately **45ps**.
- **Extracted Layout (PEX) Delay:** Following physical layout, the PEX simulation revealed that the same transition required approximately **73ps**.

3 Challenges & Observations

Testbench Design in 6T SRAM

Using basic, ideal voltage pulses in the simulator failed for the read operation because the cell's tiny transistors couldn't fight an ideal infinite source. We had to use initial condition and forced to do a read operation as the 1st operations to be done, then in order to redo read operation later we used the NMOS transistors to for the bitlines (BL/BLB) to be high but since we using NMOS so they will not charge till they reach VDD value but it's not a problem as the read operation we focus on the change that happen to the bitlines.

To prevent unrealistic instant discharging caused by the isolated single-cell testbench, 50fF dummy capacitors were added to the bitlines, successfully anchoring the voltage to reveal the true read-disturb bump.

4 Conclusion

The 6T Baseline: The 6T cell is area-efficient but vulnerable during read operations. Our simulations captured a ~220mV read-disturb bump on the internal storage node. While our careful transistor sizing prevented the cell from flipping, this shared-bitline architecture remains fundamentally stressed during every read.

The 8T Solution: The 8T architecture completely solves the read-disturb problem by adding a dedicated 2-transistor read port. Because the read path is electrically isolated from the storage nodes, our read simulation showed the internal node staying perfectly flat at 1.2V. This yields zero read noise and exceptional cell stability.

Post-Layout Extraction: proved the severe impact of physical wiring on deep-submicron designs, as the 8T internal write delay significantly degraded from 45ps in the ideal schematic to 73ps post-layout due to localized metal routing and contact capacitance. While for 6T internal write delay significantly degraded from 50ps in the ideal schematic to 92ps post-layout due to localized metal routing and contact capacitance